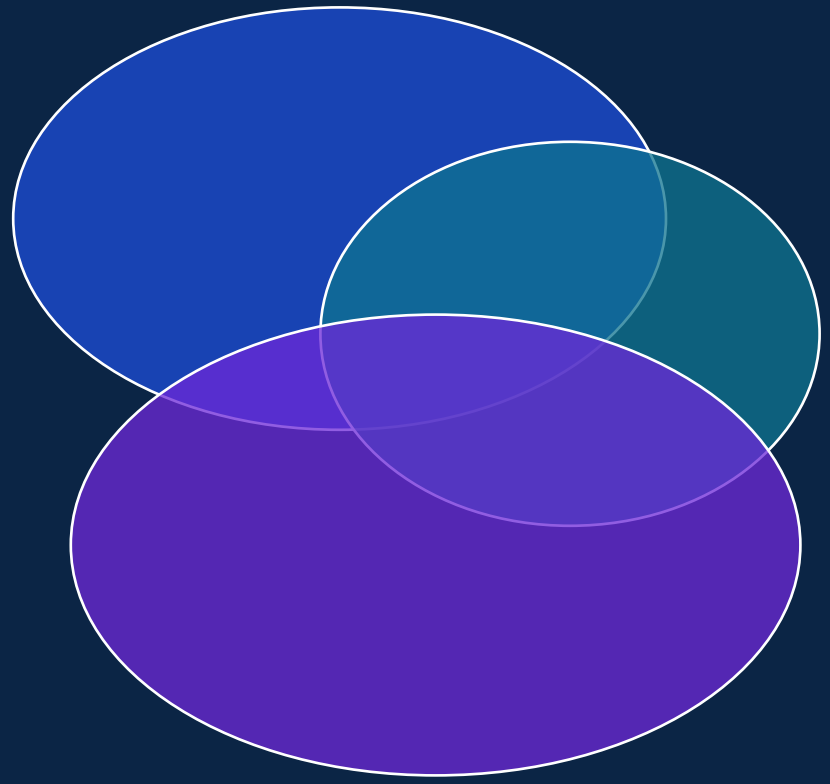
Unit:4B

Cloud Service Providers

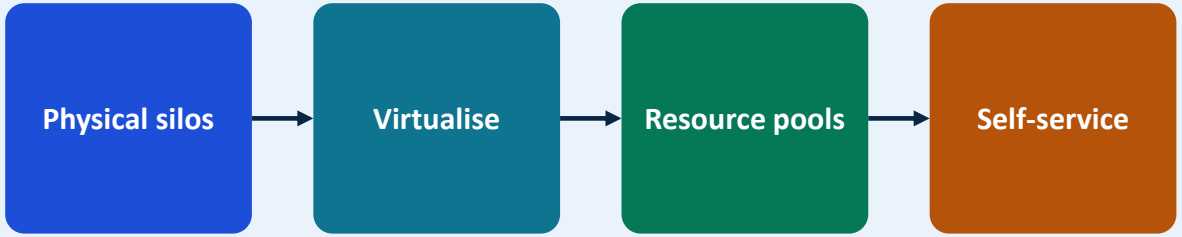


Based on Essentials of Cloud Computing by K. Chandrasekaran

EMC IT: Data Center Transformation

Legacy hardware silos become unified, software-driven resource pools.

- Move from physical silos to virtual resource pools
- Decouple workloads from storage arrays
- Automate provisioning to raise utilisation
- Isolate tenants securely across global clusters



EMC IT consolidated scattered data centres and standalone servers onto a virtualised private cloud. Applications now draw compute and storage from shared pools, provisioned through self-service portals instead of manual hardware requests. The result is higher utilisation, faster delivery to business units and lower power and space costs.

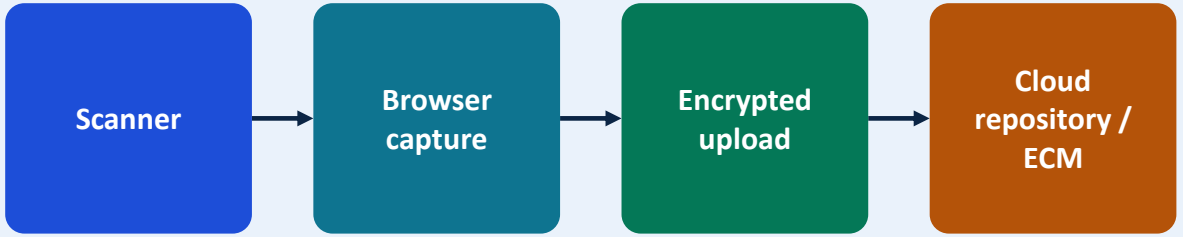
Example: A business unit requests a test environment through the portal and receives virtual machines in minutes, instead of waiting weeks for physical servers.

Note: EMC's own IT journey is the book's case study of a private-cloud transformation.

Captiva Cloud Toolkit

Browser-based document capture straight into cloud repositories.

- Scan from a standard browser, no thick client
- No local scanner driver installs
- Fast batch scanning over web protocols
- Image clean-up and enhancement tools



Traditional capture software needs installed clients and scanner drivers on every PC. Captiva Cloud Toolkit moves capture into the browser: a web page controls the scanner, enhances the image and uploads it securely to a repository. This suits branch offices and remote staff, where installing software everywhere is costly.

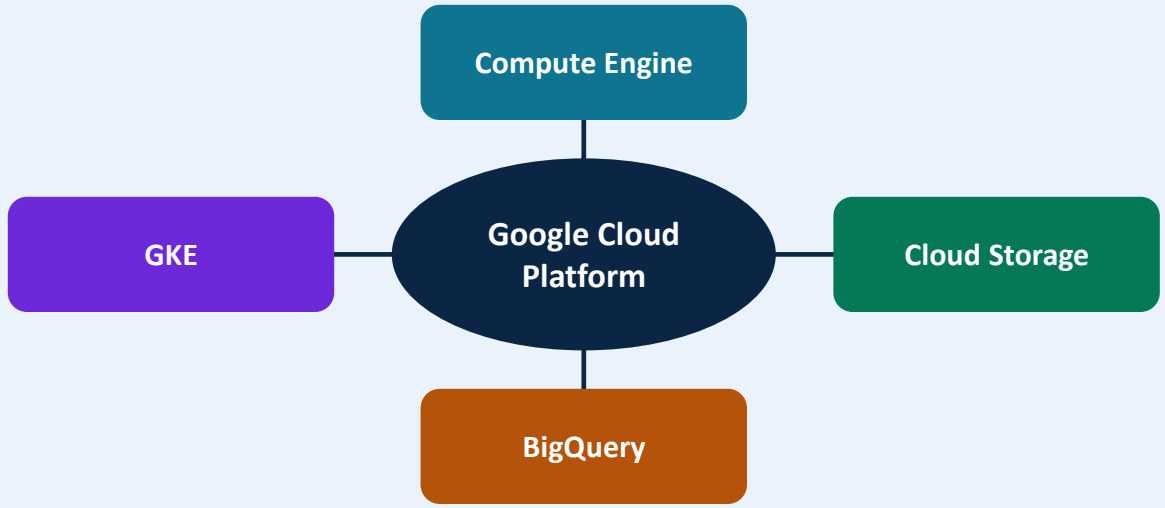
Example: A bank branch scans loan applications through a browser; the images land directly in the central content-management system with nothing installed on the PC.

Key idea: capture moves to the browser, so deployment and TCO drop.

Google Cloud Platform

Global compute, networking and analytics on Google's infrastructure.

- High-performance compute with flexible sizes
- Redundant global network and regions
- ML and big data analytics services
- IAM controls and GKE containers



Google Cloud Platform lets customers run on the same kind of infrastructure that powers Google's own services. Compute Engine supplies virtual machines, Cloud Storage holds data, BigQuery analyses it and GKE runs containers. Billing is per use, and long-running instances earn automatic sustained-use discounts.

Example: A retailer keeps sales logs in Cloud Storage and queries billions of rows in BigQuery to spot seasonal trends.

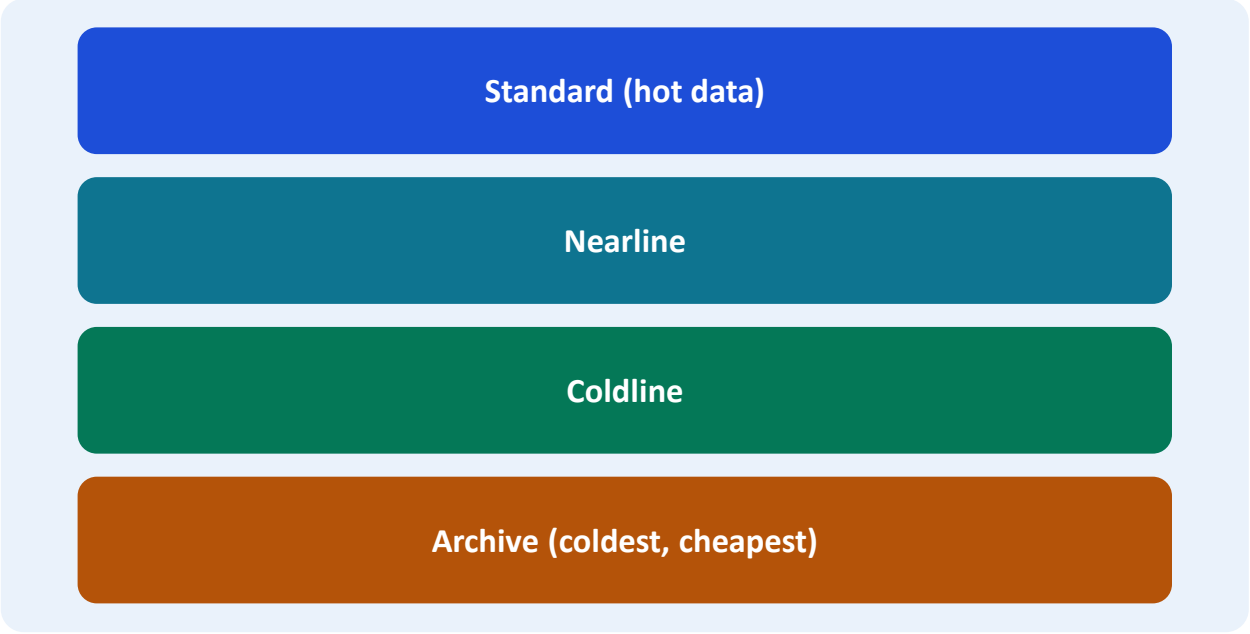
Example: Group Google's offerings by type: compute, storage, analytics, containers.

Google Cloud Storage

Durable object storage with lifecycle management.

- Object storage for unstructured data
- Multi-region, dual-region and regional options
- Lifecycle rules move data to cheaper classes
- Encrypted at rest and in transit

Data is stored as objects in buckets, with the storage class chosen by how often it is accessed. Lifecycle rules can move objects from Standard to colder classes automatically, or delete them after a set age. Versioning and default encryption protect the data, and BigQuery can analyse it directly.

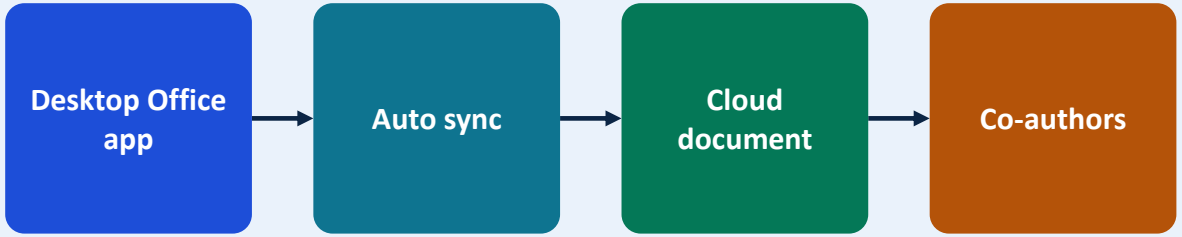


Example: Website images stay in Standard; monthly backups move to Nearline after 30 days and to Archive after a year.

Google Cloud Connect

Desktop Office suites linked to cloud storage for co-authoring.

- Connects desktop suites to cloud repositories
- Real-time multi-user co-authoring
- Automatic background sync and version history
- Offline edits resolved after sync



Cloud Connect added a plug-in to desktop Office applications so that a document could be saved to the cloud and edited by several people. Changes sync in the background and each revision is kept in the history. It showed how desktop software and cloud collaboration could be combined.

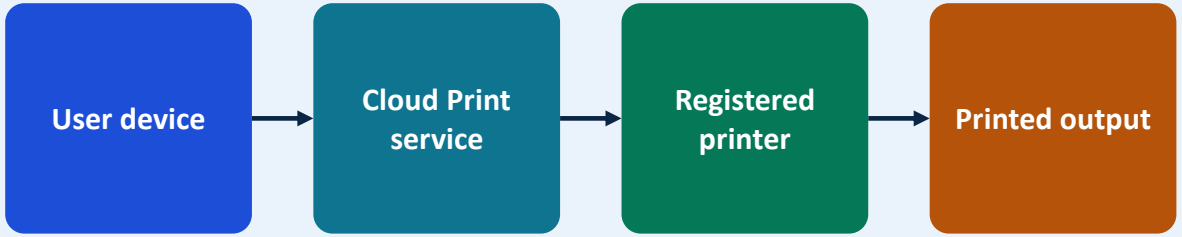
Example: Three team members edit one Word report; each saves, the changes merge, and anyone can roll back to an earlier version.

Note: Cloud Connect is a book-era product; today Google Workspace covers this.

Google Cloud Print

Printers registered to the cloud, printable from any device.

- Connect home and office printers to the cloud
- Print from phones and remote locations
- No printer driver installs
- Share printers with authorised users



A printer is registered with the Cloud Print service, which gives it a cloud identity. Users send a job from any device, the service queues it, and the printer collects it over a secure connection. No drivers are needed on the sender's device, and printers can be shared by permission.

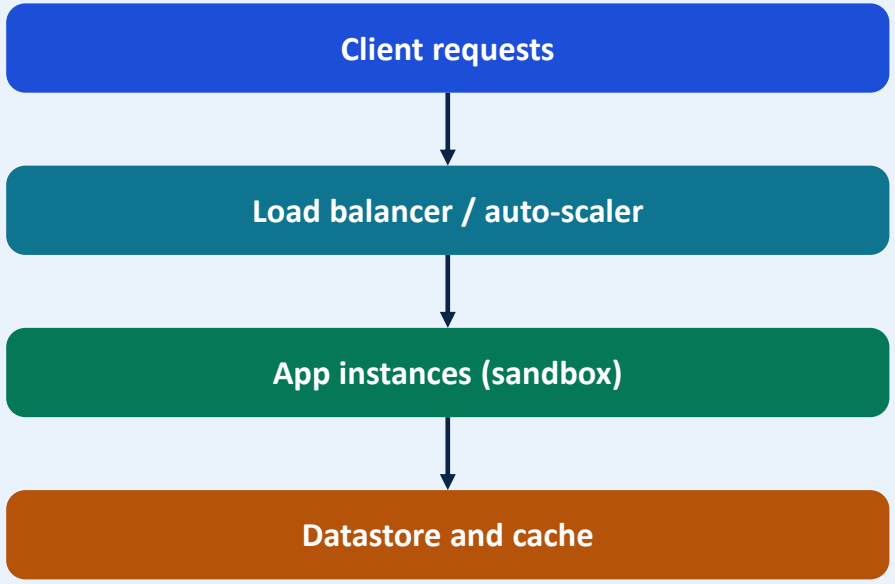
Example: A student prints from a phone to the library printer without being on the campus network.

Note: Cloud Print was retired by Google at the end of 2020.

Google App Engine (PaaS)

Fully managed platform for scalable web apps.

- Build apps without managing servers
- Automatic horizontal scaling
- Runtimes for popular languages
- Built-in datastore and caching



Developers upload code and App Engine handles everything else: provisioning instances, load balancing, scaling up under heavy traffic and scaling down when it falls. Applications run in sandboxes, so tenants are isolated. The trade-off is that apps must follow the platform's runtimes and services, which increases lock-in.

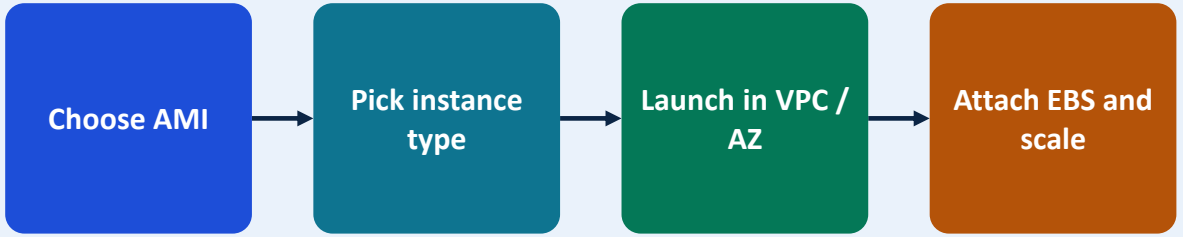
Example: A ticket-booking site gets a surge of visitors on release day; App Engine adds instances automatically and removes them afterwards.

Example: App Engine is the standard PaaS example next to Azure and Aneka.

Amazon EC2

Resizable virtual servers with full root control.

- Launch instances from Amazon Machine Images
- Scale up or down on demand
- Instance types for compute, memory, storage
- On-Demand, Spot and Reserved pricing



An EC2 instance is launched from an Amazon Machine Image, which holds the operating system and software. The customer picks an instance type, a Region and Availability Zone, and a network in a Virtual Private Cloud. EBS volumes give persistent disks, and load balancing with Auto Scaling keeps applications available.

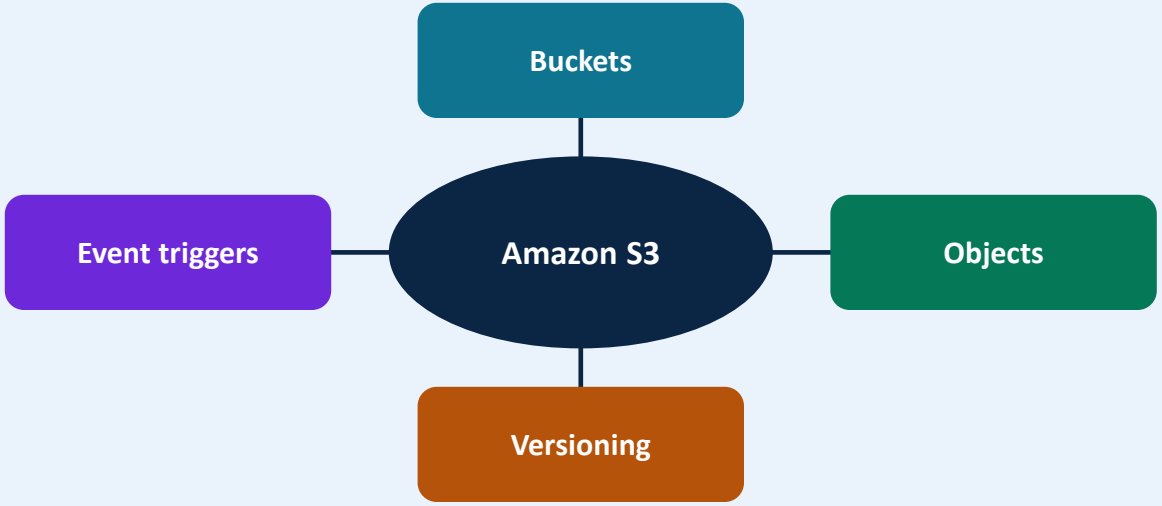
Example: A start-up runs its web server on On-Demand instances and uses Spot instances for overnight batch jobs to save cost.

Note: EC2 is IaaS: you manage the OS and above.

Amazon S3

Web-scale object storage with 11-nines durability.

- Buckets hold objects of any type
- 99.999999999% designed durability
- Bucket policies, ACLs and encryption
- Versioning, object lock, cross-region replication



S3 stores data as objects, each identified by a key inside a bucket. Data is replicated across multiple facilities, giving 99.999999999% designed durability. Access is controlled by bucket policies and ACLs, and tiering features reduce cost. S3 can also host static websites and trigger functions on upload.

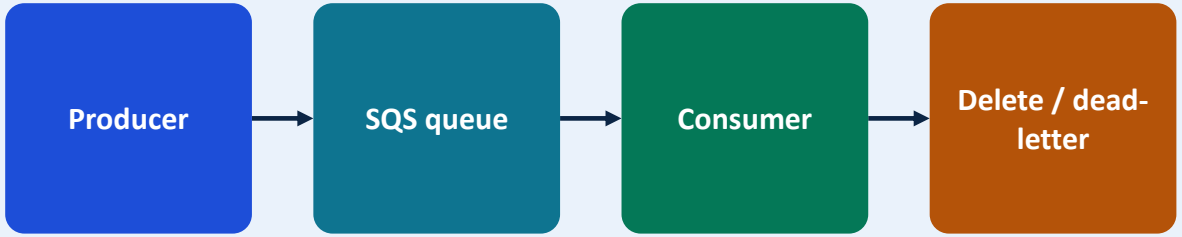
Example: A photo app stores each upload in S3; an upload event triggers a function that creates thumbnails.

Remember: bucket, object, key. S3 is object storage, not a file system.

Amazon SQS

Managed queues that decouple application components.

- Fully managed message queuing
- Standard queues for throughput, FIFO for order
- Dead-letter queues isolate failures
- Visibility timeout prevents duplicate processing



In a tightly coupled system, a slow component blocks the one that calls it. SQS places a queue between them: the producer sends messages and consumers read them at their own pace. A message stays hidden during its visibility timeout, and repeated failures move to a dead-letter queue for inspection.

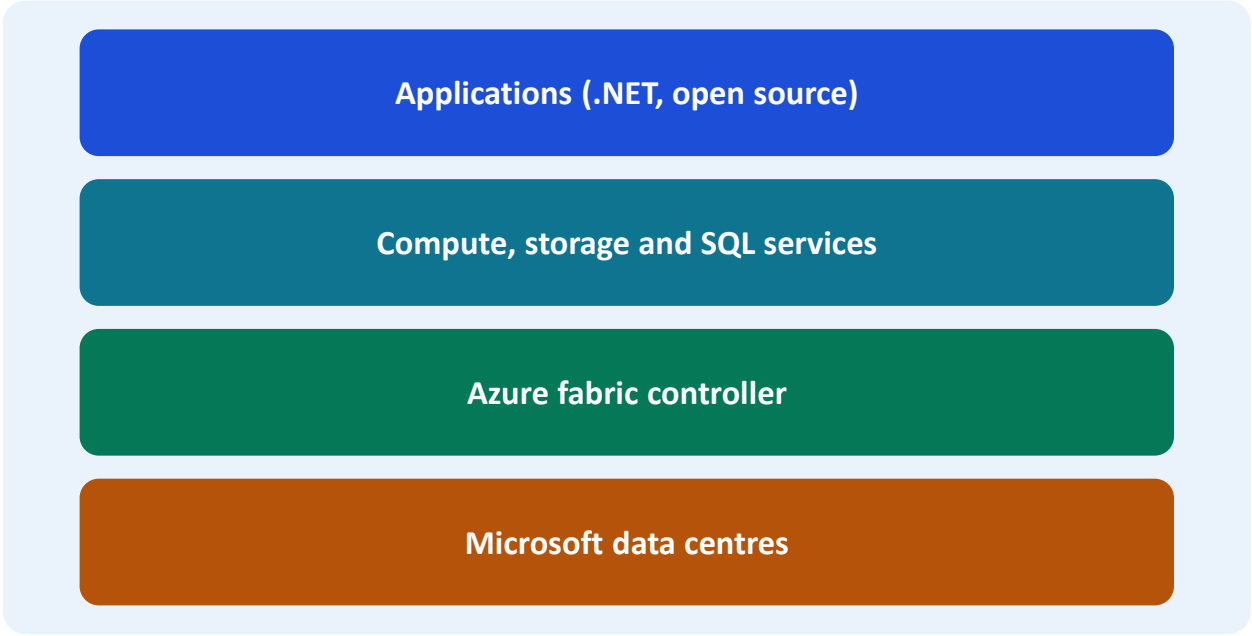
Example: An online shop places each order in a queue; the payment, stock and email services process orders independently.

Example: SQS gives asynchronous, loosely coupled communication.

Windows Azure

Microsoft's cloud platform combining IaaS and PaaS.

- Cloud operating system for hosted applications
- Supports .NET and open-source stacks
- Relational and non-relational data services
- Azure Active Directory for identity



Azure runs applications in Microsoft data centres, managed by the fabric controller, which allocates virtual machines and monitors their health. It offers PaaS roles for web and worker processes as well as IaaS virtual machines. Integration with Active Directory and Windows tools appeals to enterprises.

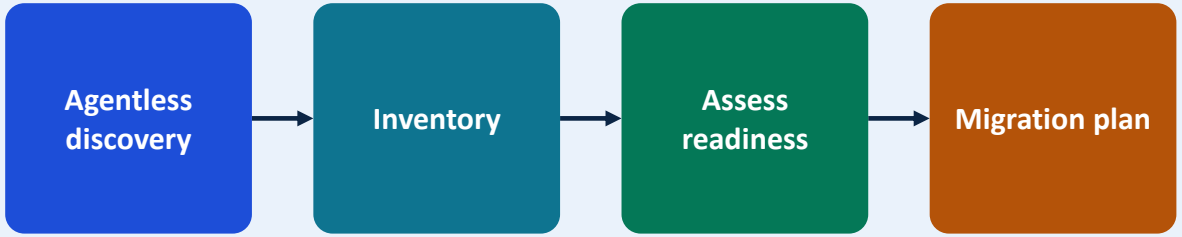
Example: A company hosts its ASP.NET web application as an Azure web role and keeps its records in Azure SQL Database.

Example: Windows Azure is now called Microsoft Azure.

Microsoft MAP Toolkit

Agentless assessment before migration or consolidation.

- Discovers and inventories IT assets
- Checks readiness for Azure or Hyper-V
- Analyses server utilisation
- Tracks software license usage



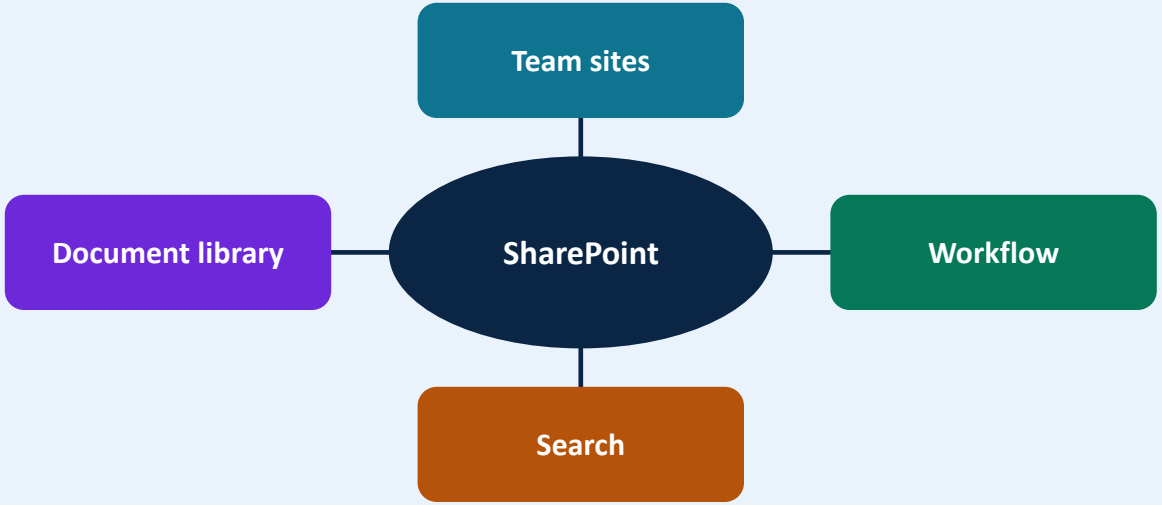
Before moving to the cloud, an organisation needs to know what it owns. The MAP Toolkit scans the network without agents, collects hardware, software and utilisation data, and reports which servers are ready for Azure or virtualisation. It also supports consolidation planning and license compliance checks.

Example: An IT team scans its servers and finds many running at very low CPU use, making them good candidates for consolidation onto fewer hosts.

Microsoft SharePoint

Collaboration, intranet and document management.

- Team sites for sharing and tracking
- Enterprise search across repositories
- Workflow automation
- Permission hierarchies and co-authoring



SharePoint provides a web portal where teams keep documents, calendars and tasks together. Document libraries support check-in, versioning and permissions, while workflows route items for approval. Because it is browser-based and integrates with Office, it became a common intranet and collaboration platform.

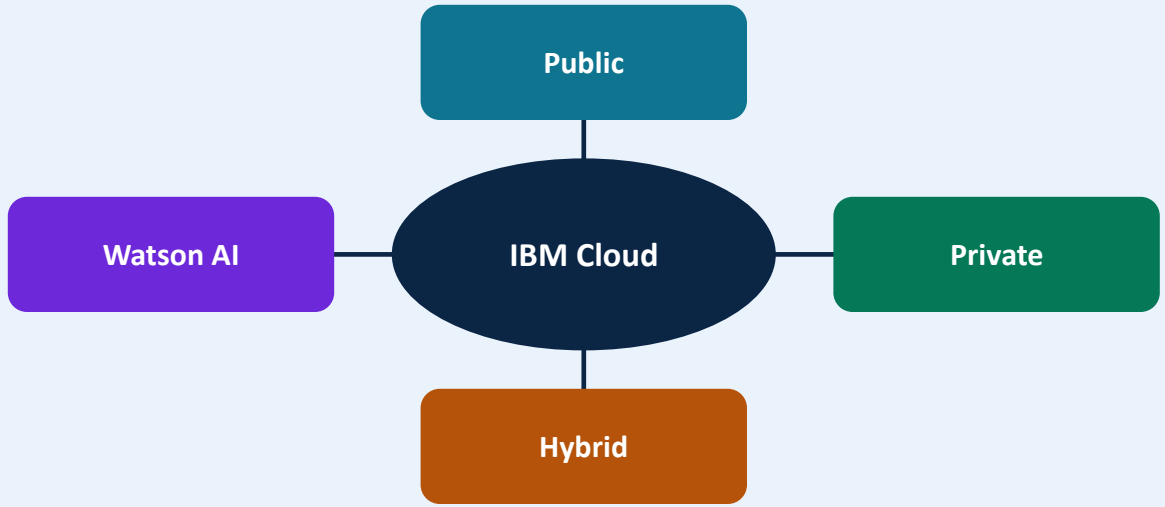
Example: A project team keeps its plan in a SharePoint library; a manager approves each revision through a workflow.

Example: SharePoint is delivered as SaaS-style collaboration (Online) or on-premises.

IBM Cloud Models

Public, private and hybrid enterprise delivery.

- Public, private and hybrid deployment models
- Focus on security, governance and compliance
- Watson AI inside cloud workloads
- Bare metal and virtual servers



IBM positions its cloud for large enterprises that must combine existing systems with new services. Customers can choose public, private or hybrid deployments, with governance and security built in. Watson AI services and OpenShift containers let organisations modernise applications gradually rather than rewrite them.

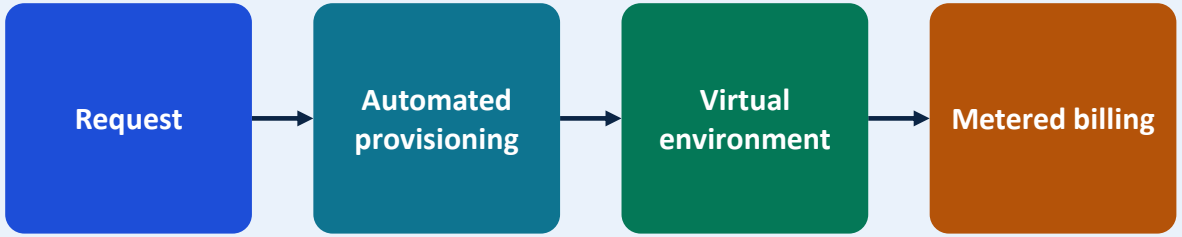
Example: A hospital keeps patient records in a private cloud and uses public cloud capacity for research analytics.

Example: Linking IBM to enterprise hybrid cloud and governance.

IBM SmartCloud

Enterprise infrastructure with automated provisioning.

- High-performance enterprise computing
- Automated provisioning of virtual environments
- Isolation for multi-tenant security
- Backup, recovery and resilience



SmartCloud offered enterprise-class infrastructure with self-service requests and automated provisioning of virtual machines. Resources were metered and billed by use, with isolation between tenants and built-in backup and recovery. It combined IBM's utility-computing experience with cloud delivery.

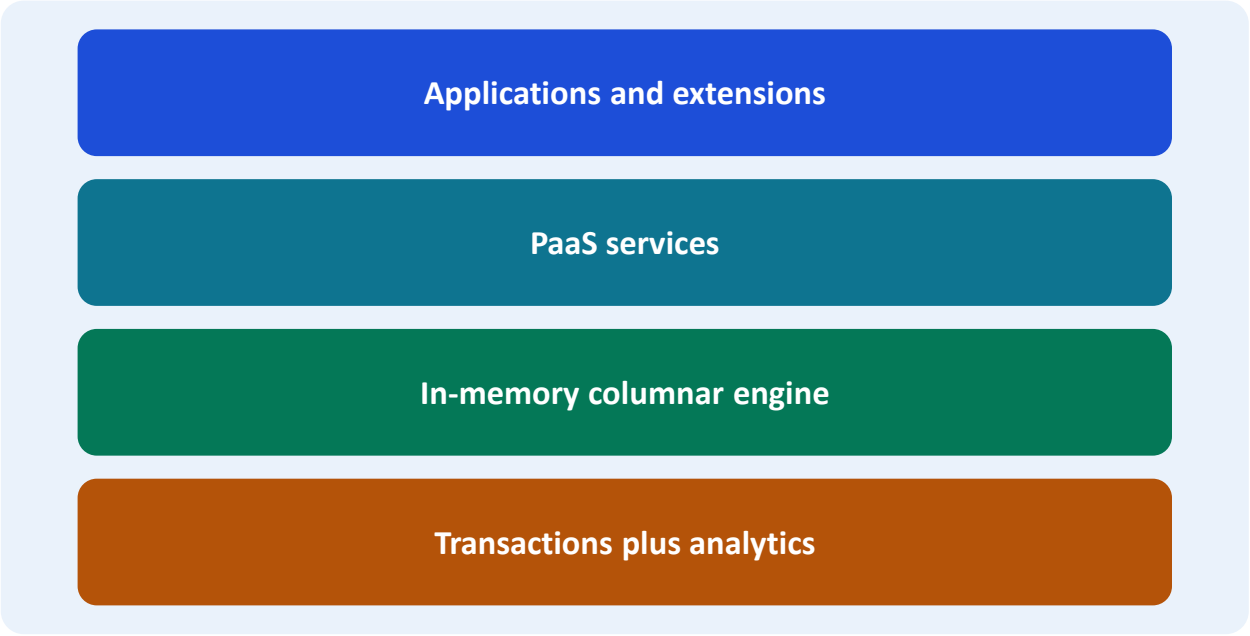
Example: A development team requests ten virtual machines for a month and is billed only for the hours they run.

Example: SmartCloud is IBM's earlier brand; it now sits under IBM Cloud.

SAP HANA Cloud Platform

In-memory database plus PaaS for real-time enterprise apps.

- In-memory database and application platform
- Real-time analytics and transactions together
- Rapid cloud-native extensions
- Predictive analytics and ML services



Conventional systems keep transactions in one database and analytics in another, copied across overnight. HANA holds data in memory in a columnar format, so both workloads run on the same data in real time. The cloud platform adds PaaS services for building extensions to SAP business applications.

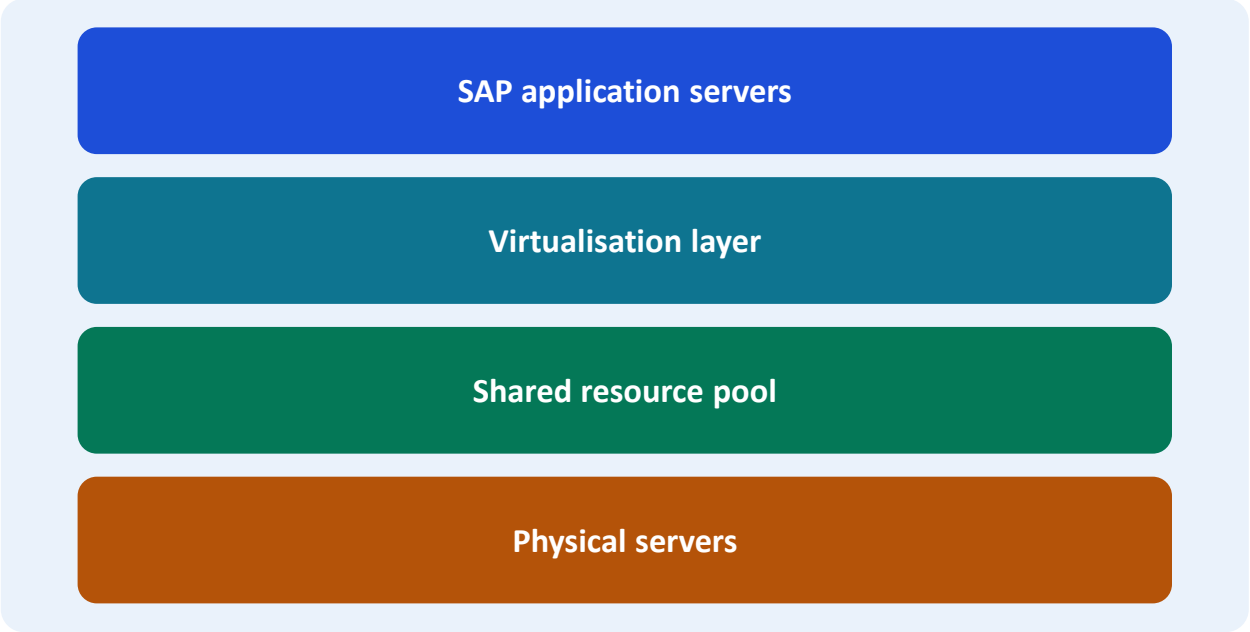
Example: A manufacturer checks live sales and stock levels on the same system that records each order.

SAP Virtualization Services

Enterprise software decoupled from physical hardware.

- Virtualise SAP application servers
- Migrate workloads across resource pools
- Automated high availability and failover
- Fast provisioning and landscape cloning

SAP landscapes include many application servers, databases and test systems. Running them on virtual machines allows a system to be cloned in minutes, moved to another host without downtime and restarted automatically after a failure. Fewer physical servers also mean lower energy and space costs.

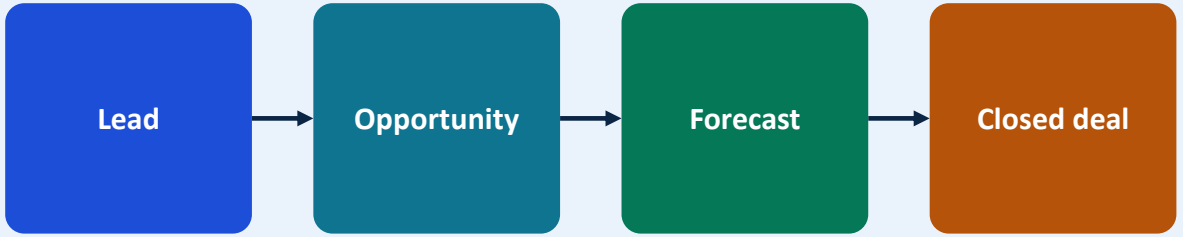


Example: An administrator clones the production landscape to build a test system for an upgrade, then deletes it afterwards.

Salesforce Sales Cloud

Multi-tenant cloud CRM with pipeline automation.

- Cloud customer relationship management
- Automates leads and opportunities
- Forecasting and analytics dashboards
- Mobile access for field sales



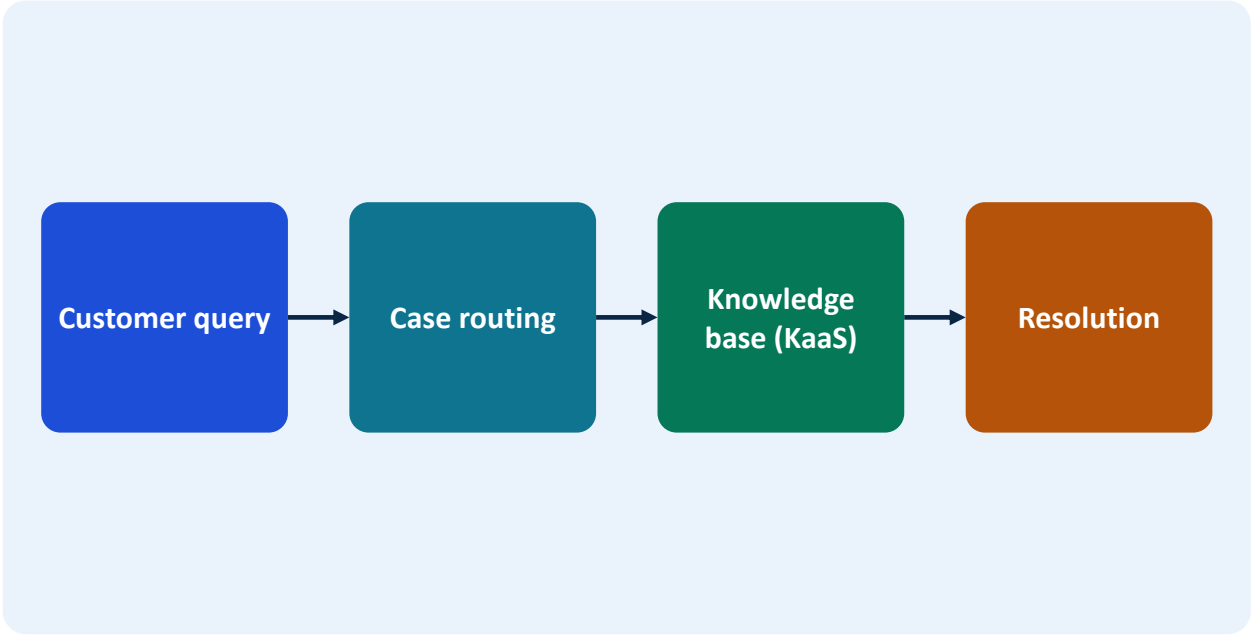
Sales Cloud is delivered as SaaS on a multi-tenant platform: all customers share the same infrastructure and code, while their data stays separate. Sales staff record leads, track opportunities through stages and see forecasts on dashboards, from a browser or phone. Add-ons come from the AppExchange marketplace.

Example: A representative logs a lead from a trade show and moves it to 'Proposal' after a meeting; the manager's forecast updates automatically.

Example: Salesforce is the classic SaaS .

Omni-channel support with Knowledge as a Service.

- Customer service and case management
- Knowledge as a Service (KaaS) resolves issues faster
- Routes cases from email, chat and phone
- Self-service portals and communities



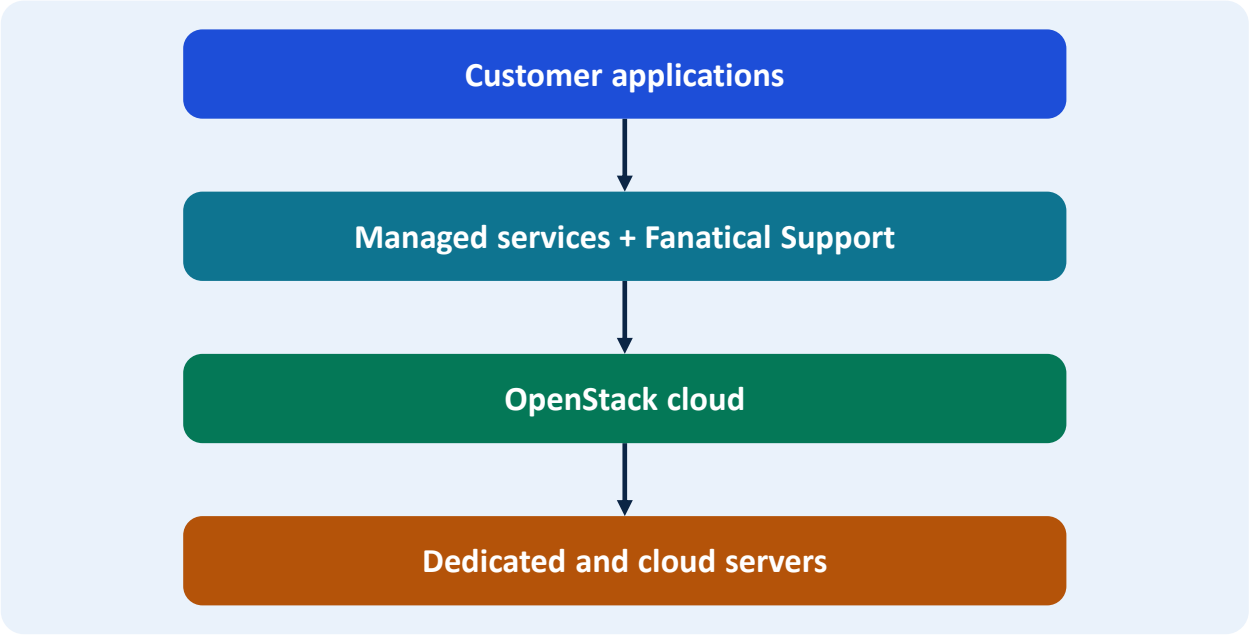
Service Cloud brings customer queries from email, chat, phone and social media into one console. Cases are routed to the right agent, who is supported by a searchable knowledge base. Knowledge as a Service means solutions written once are shared with agents and customers, so repeated problems are solved faster.

Example: A customer asks how to reset a password; the portal shows a knowledge article and no support call is needed.

Rackspace

Managed multi-cloud with Fanatical Support.

- Managed services across multi-cloud
- Expert 'Fanatical Support'
- OpenStack-based open cloud
- Dedicated servers plus cloud nodes



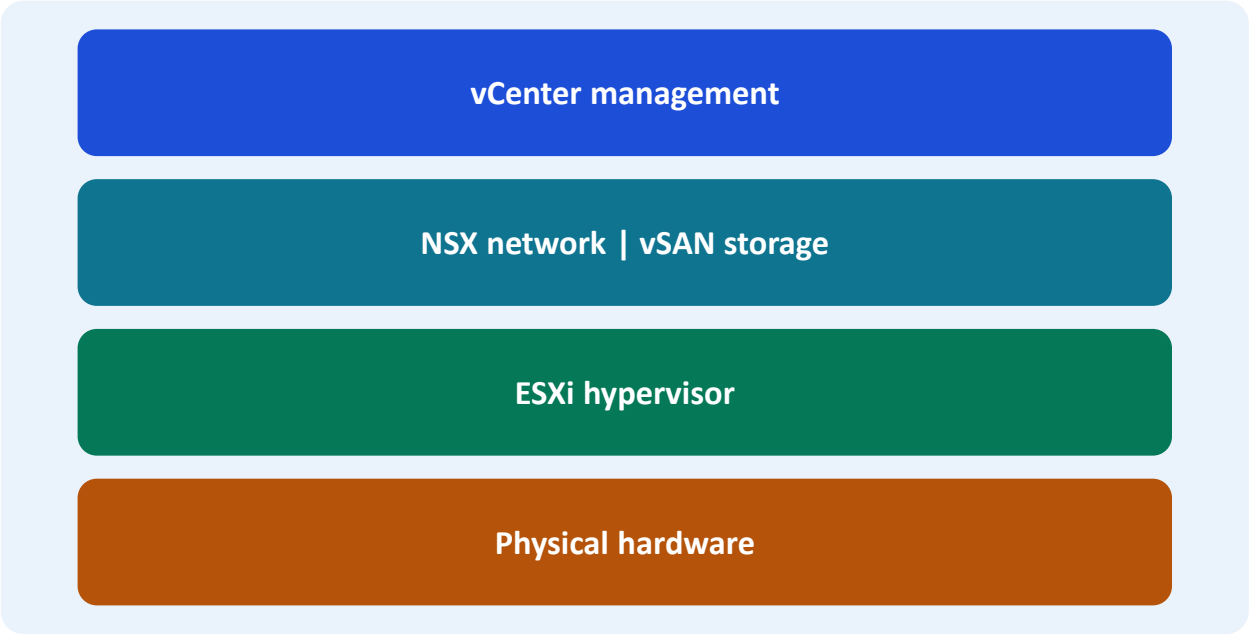
Rackspace sells expertise as much as infrastructure. It runs customers' servers in its own data centres, on OpenStack clouds or on other providers' platforms, and takes responsibility for monitoring, patching and security. Smaller firms without large IT teams use it to gain cloud benefits with less internal effort.

Example: A small retailer lets Rackspace manage its web servers and database around the clock instead of hiring a systems team.

VMware

Virtualisation leader building the software-defined data centre.

- ESXi hypervisor pioneered server virtualisation
- SDDC combines compute, network and storage
- NSX network and vSAN storage virtualisation
- VMware Cloud enables hybrid workload mobility



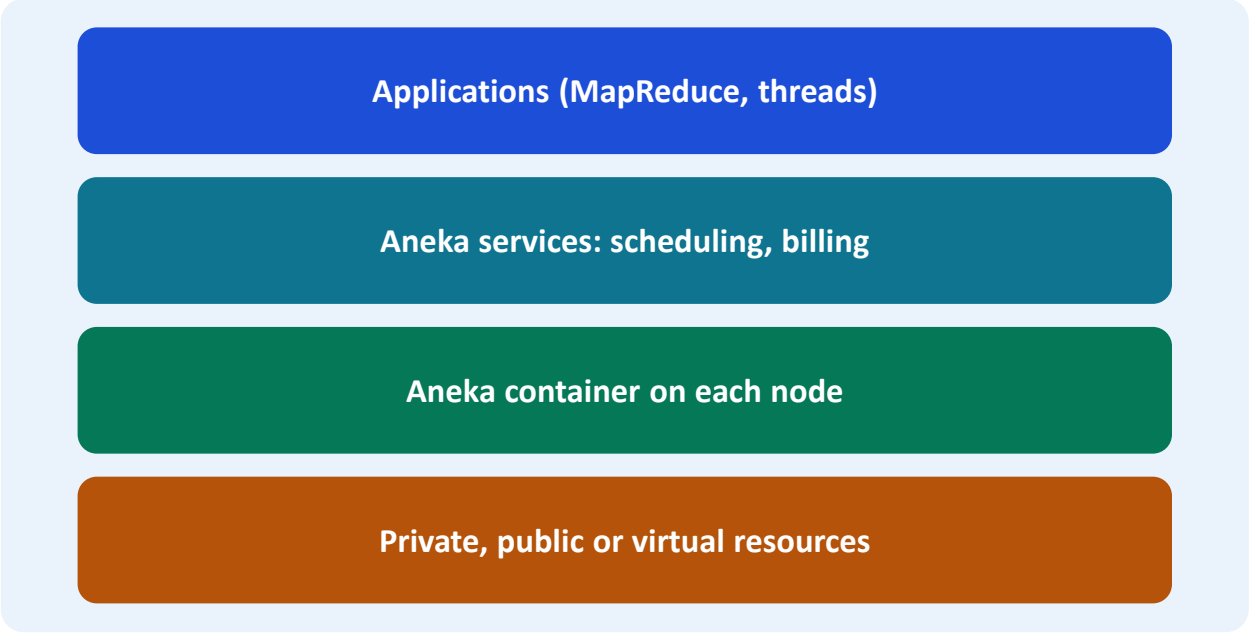
ESXi is a bare-metal hypervisor that lets many virtual machines share one physical server. VMware extended this to the software-defined data centre by virtualising networking with NSX and storage with vSAN, all managed through vCenter. Live migration moves running machines between hosts, supporting hybrid mobility.

Example: vMotion moves a running database VM to another host so the first host can be maintained with no downtime.

Manjrasoft Aneka

PaaS framework for private and public clouds.

- Platform for building and managing clouds
- Multiple programming models, such as MapReduce and threads
- Billing and accounting modules
- Dynamic provisioning and scheduling



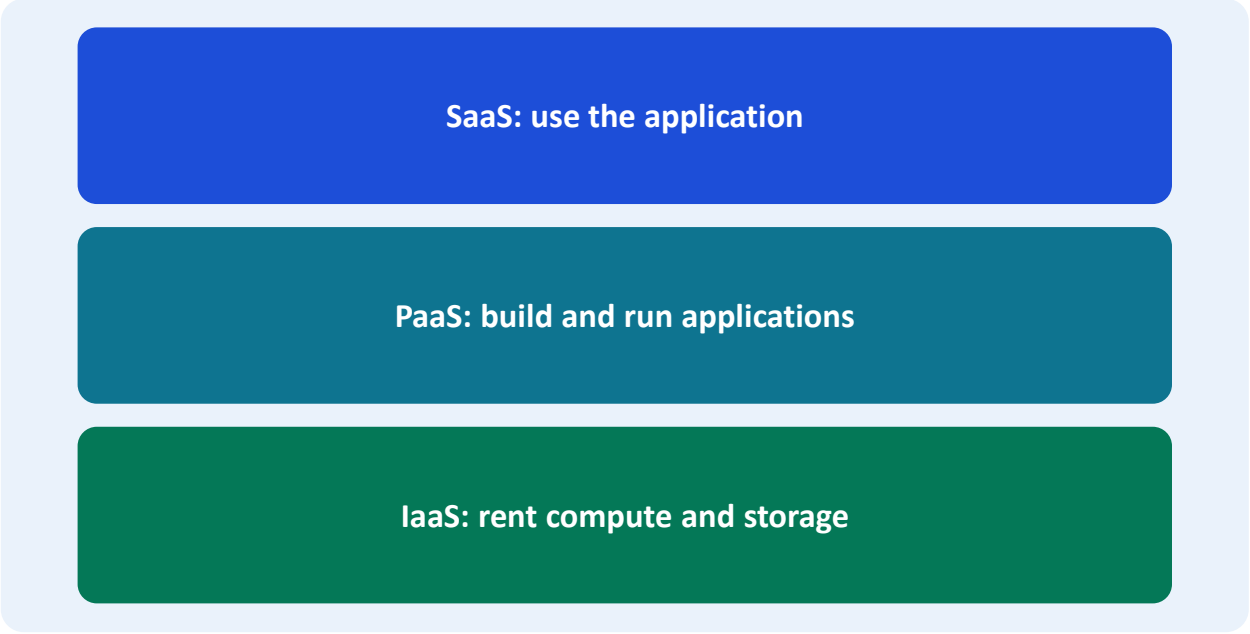
Aneka is a .NET-based platform that grew out of research at the University of Melbourne and is commercialised by Manjrasoft. Applications use programming models such as threads, tasks or MapReduce, and Aneka's scheduler spreads work across nodes. Accounting modules measure usage, so it can run a private cloud or burst into a public one.

Example: A research lab runs a MapReduce analysis on idle desktop PCs and extends to public cloud servers when a deadline nears.

Comparing IaaS, PaaS and SaaS

Responsibility shifts from client to provider as you go up.

- IaaS: rent servers, manage OS and apps (EC2)
- PaaS: deploy code on a platform (App Engine, Azure)
- SaaS: use ready software (Salesforce, SharePoint)
- Higher level means faster start but more lock-in



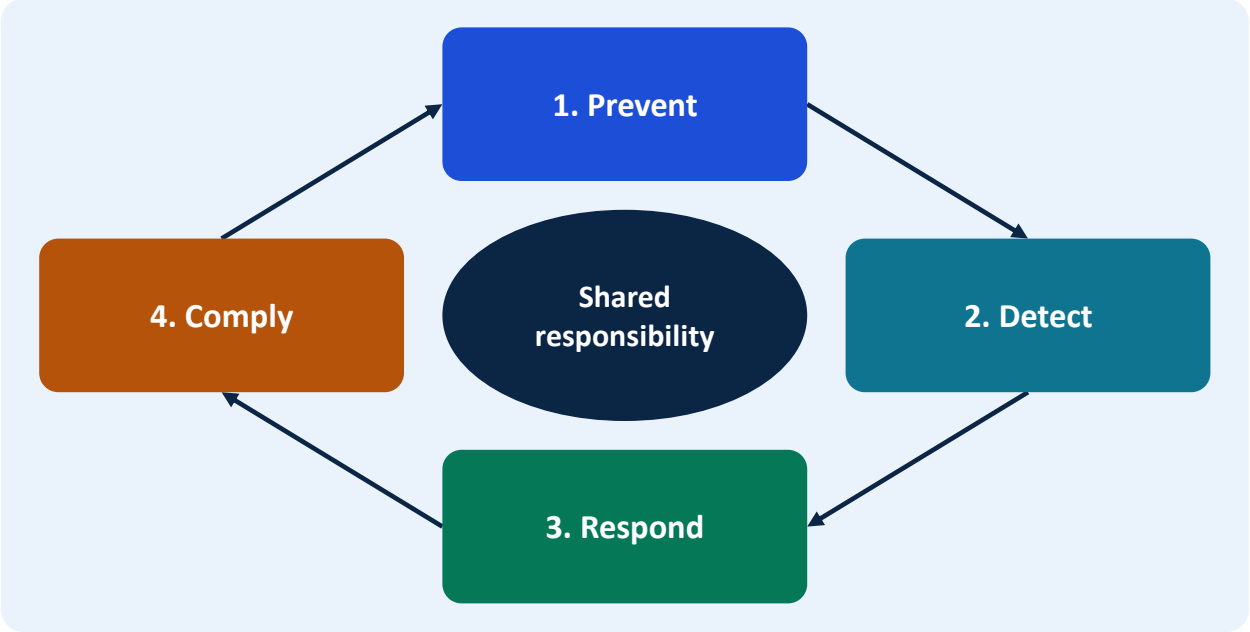
The three models differ in how much the customer manages. With IaaS the customer controls the operating system and software; with PaaS only the application and data; with SaaS only the use of the service. Moving up the stack speeds delivery and reduces effort but gives less control and more dependence on the vendor.

Example: Renting an EC2 server is IaaS, deploying code on App Engine is PaaS, and using Salesforce in a browser is SaaS.

Security & Compliance Standards

Layered controls protect tenant workloads.

- Encryption at rest and in transit
- Multi-factor authentication and IAM
- GDPR, HIPAA and SOC 2 compliance
- Vulnerability scanning, logging and intrusion detection



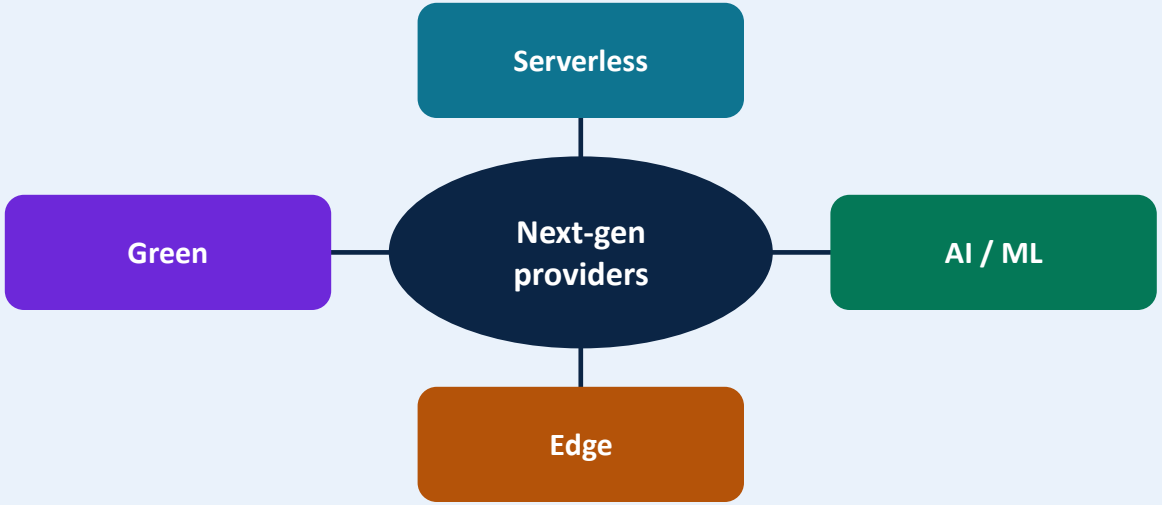
Security is a shared responsibility. The provider protects facilities, network and the virtualisation layer, while the customer manages identities, data and application settings. Strong authentication, encryption and logging reduce risk, and audits such as SOC 2 and regulations like GDPR and HIPAA show that controls are verified.

Example: A hospital stores patient data in the cloud encrypted with its own keys, with MFA for staff and audit logs for HIPAA reviews.

Future Provider Trends

Serverless, AI, edge and green data centres shape the next generation.

- Serverless, event-driven architectures
- AI and ML built into services
- Multi-cloud and hybrid interoperability
- Edge nodes for IoT latency



Providers are moving towards services where developers manage less infrastructure. Serverless platforms run code on events, AI services are built into storage and analytics, and edge nodes process data near devices. Data centres are being designed for lower energy use, and workloads should move freely between clouds.

Example: A smart-city camera analyses traffic at an edge node and sends only summaries to the cloud.